

PILOT COMBAT FLIGHT COURSE MANUAL



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AERIAL TRAINING COMMAND

Kali Special Operations Legion, Headquarters

Foreword by the Commander

1. PURPOSE

Document KSOL-PF 6-1A (Pilot Combat Flight Course Manual) establishes the fundamental knowledge, terminology, procedures, and practical skills required for personnel to operate rotary-wing aircraft in support of KSOL activities and operations within BRM5.

The course is intended to develop competent pilots capable of controlling an aircraft, performing basic flight and landing manoeuvres, communicating effectively with ground elements, and safely integrating aviation assets into an operational element.

2. SCOPE

This document covers the fundamentals of helicopter flight, aircraft controls, helicopter movement, basic flight manoeuvres, take-off and landing procedures, landing types, aircraft handling, flight terminology, tactical aviation fundamentals, passenger and ground-element coordination, and standardised aviation radio communications.

General, Kali Special Operations Legion
Commander
'Obi' Aka Vandam
Special Insertion Service, KSOL

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APPENDIX 'A'

Helicopter Fundamentals

AA01. Rotary-Wing Aircraft

A helicopter is a rotary-wing aircraft which produces lift from one or more rotating rotor system.

**Through out the document "aircraft" is simply referring to a rotary-wing aircraft*

AA02. Main Aircraft Controls

AA02 A. Cyclic

The cyclic of a helicopter controls forward/backward movement (W/S key on keyboard), called "**Pitch** " and also left/right (A/D) movement called "**Roll**". Cyclic is controlled by the lever in front of the pilot.

AA02 B. Collective

The collective of a helicopter controls vertical lift from the main rotor blades. **Collective up/down** moves the helicopter vertically up/down.

AA02 C. Pedals

Pedals make the helicopet **yaw** on its vertical axis or the 2-D plane.

AA03. Understanding Pitch

Pitch Positive means pulling the **nose up** making the the helicopter reduce forward speed and coming to stall at a point, or to gain climb lift, depending on the helicopter's state.

Pitch Negative means pulling the **nose down** making the helicopter to gain forward speed, or to descend rapidly towards the ground, depending on the helicopter's state.

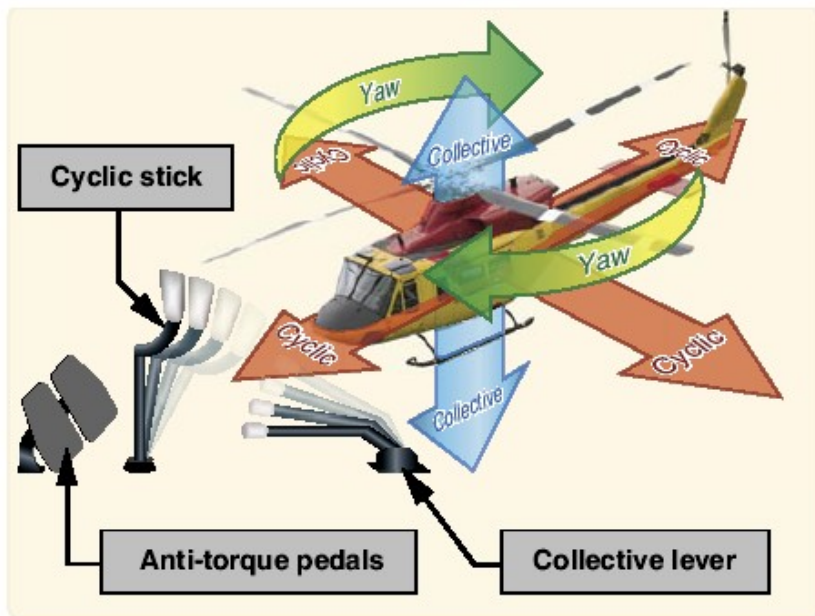


Figure 5-3. Helicopter flight control system.

APPENDIX 'B'

Basic Helicopter Movement

BA04. Hover

Hover is a flight when the aircraft maintains same stable horizontal and vertical position & altitude.

For a stable hover, the aircraft requires small inputs of: Cyclic, yaw, and collective from the pilot in control.

BA05. Vertical Movement

Vertical movement is primarily controlled by the collective lever.

BA05 A. Climb - Increase collective as required.

BA05 B. Descend – Reduce collective as required.

BA06. Forward/Reverse Flight

Forward flight is primarily controlled by cyclic forward input.

Tilts Forward > Redirects Rotor Thrust > Accelerates Forward > Requires appropriate power management

Once the desired flight attitude is established, excessive cyclic input should be avoided.

For **Reverse flight** cyclic backward input is primary. Consequences are vice-versa to forward flight.

BA07. Lateral Flight

Lateral flight moves the aircraft sideways without requiring a corresponding change in heading.

Left Lateral Movement

Left cyclic.

Right Lateral Movement

Right cyclic.

Lateral movement is particularly useful for:

- Positioning.
- Hover adjustments.
- Landing alignment.
- Formation spacing.
- Confined-area manoeuvring.

APPENDIX 'C'

Basic Flight Manoeuvres

CB08. Take Off

A basic helicopter take-off consists of:

1. Establishing aircraft control.
2. Increasing collective.
3. Becoming airborne.
4. Establishing the required attitude.
5. Accelerating into the desired direction.
6. Establishing the required altitude and heading.

The pilot should avoid unnecessary control movements during the transition from hover to forward flight.

CB09. Climb

A climb increases aircraft altitude.

The pilot should:

- Maintain directional control.
- Establish appropriate climb attitude.
- Maintain awareness of surrounding terrain.
- Avoid excessive airspeed or altitude changes.

CB10. Descent

A descent reduces altitude.

The pilot should:

- Maintain control.
- Monitor ground clearance.
- Maintain appropriate speed.
- Establish the intended landing or flight path.

CB11. Level Flight

In a level flight pilot maintains, stable altitude, heading, and airspeed.

CB12. Orbit

Orbit means flying the aircraft in a controlled circular flight path around a reference point. It can be used to observe, provide aircover/overwatch, or coordination.

Pilot should be aware of: ground situation, altitude, path, airspeed, reference point and threats from air or ground.

CB13. Approach

Approach is the controlled flight heading path towards an LZ (Landing Zone).

APPENDIX 'D'

Landing Procedures

DC14. Landing Fundamentals

A landing consists of transitioning from flight to ground contact while maintaining control of:

- Position.
- Heading.
- Altitude.
- Descent.
- Speed.

The pilot should prioritise **controlled aircraft movement over speed of execution.**

DC15. Normal Landing

Normal landing is when conditions are under routine with sufficient visibility, and space.

Standard sequence:

Approach > Deaccelerate > Descend > Hover/Settle > Touchdown

Aircraft to be aligned along intended Landing area.

DC16. Vertical Landing

A vertical landing involves descending primarily downward into a suitable landing area.

It is useful when:

- Space is limited.
- Forward movement is undesirable.
- The pilot is already established over the landing point.

DC17. Running Landing

This includes rapid descending and deaccelerating the helicopter. This is a situational and conditional landing. Depends primarily on pilot's expertise and given condition.

DC18. Confined Area Landing

A confined-area landing is performed where the available landing area is limited by:

- Buildings.
- Trees.
- Terrain.
- Structures.
- Other obstacles.

DC19. Landing Zone (LZ)

Landing zone is a designated zone/area for an aircraft to land. LZ can be identified by grid reference/markers on map/ Laser/IR strobes/ smokes/flares.

DC20. Drop Zone (DZ)

Drop zone is a designated zone/area for paratroopers to exit out of an aircraft to glide/free-fall to their designated LZ, while the aircraft continues to fly to its designated heading.

APPENDIX 'E'

Aircraft Handling

ED21. Aircraft Position Awareness

Pilot should be aware of: **Position** (Where am I?), **Altitude** (How high am I?), **Heading** (Where am I facing?), **Movement** (Where am I moving?).

These form the base of basic flight awareness.

ED22. Situational Awareness

Pilot should be aware of:

- Hazards (terrain, obstacles, structures, ridges, tree-lines, etc).
- Ground element/situation/movement.
- Friendlies/hostiles/enemy/faker/bandits/other elements' positional awareness.
- LZ/DZ and mission objectives, and priority.

ED23. Passanger Awareness

Transport pilots must understand that their passengers are part of the mission. The pilot should know:

- Who is being transported.
- Where they are being delivered.
- When they are expected.
- Whether the landing zone is ready.
- Whether the element has successfully mounted/disembarked.

ED24. RULE OF THUMB

These are few rules & SOPs (Standard Operating Procedures) a pilot must remember and practice at all times.

- Understand that a pilot is in-charge & apex authority when aboard.
- Read the basic rules to the passenger about what they are expected to do:
 - DO NOT dismount the helicopter until the pilot says so.
 - DO NOT interfere in basic functionality and operational responsibility of the pilot & the air-crew.
- Ensure (only at night flight) to change your cabin color to neutral/green when it is time for the passengers to disembark.
- Communicate with C2/passengers/All-units about your movements/intentions/heading from time to time to avoid confusion or assumptions.
- Understand: Pilots primary objective is to safely transport troops at any cost.

APPENDIX 'F'

Basic Tactical Aviation

FE25. Pilot & Ground/C2 Element

Aviation and ground personnel operate as one mission element.

The pilot provides:

- Mobility.
- Observation.
- Transport.
- Extraction.
- Aviation support.

The ground/C2 (Command & Control) element provides:

- Mission requirements.
- Landing-zone information.
- Passenger status.
- Ground situation updates.
- Direction and coordination.

FE26. Chalk

A **Chalk** is a designated group of personnel or a designated aircraft movement associated with a transport mission.

Do not confuse it with the actual callsign of the helicopter/crew.

FE27. Aircraft Status

Status	Reference
Inbound	Moving towards designated location.
Outbound/Moving away	Moving away from designated location.
Overhead	Is above or directly over designated location.
Stand-by	Waiting for orders/tasking
Going-around	Taking a sortie circle back to the location
On station	Reached/arrived at designated position and is ready to take/execute a task.
Bingo	Out of fuel
Winchester	Out of ammunition
RTB	Return(-ing) To Base

APPENDIX 'G'

Attack Helicopters

GF28. Overview

Attack Helicopters are helicopters which are armed and are intended for attack missions/fly-bys.

GF29. Strafe Run

A **strafe run** is an attack pass in which an aircraft flies along a target area while engaging targets during the pass, usually with a forward-firing gun.

GF30. Hover Attack

The aircraft remains approximately stationary while engaging a target.

GF31. Advanced Diving Attack

The aircraft descends toward the target during the attack before breaking away

GF32. Weapons Envelope

A **weapons envelope** is the range of conditions/formation in which a particular weapon can effectively be employed against a target.

“MEN APART, EVERY MAN AN EMPEROR”

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